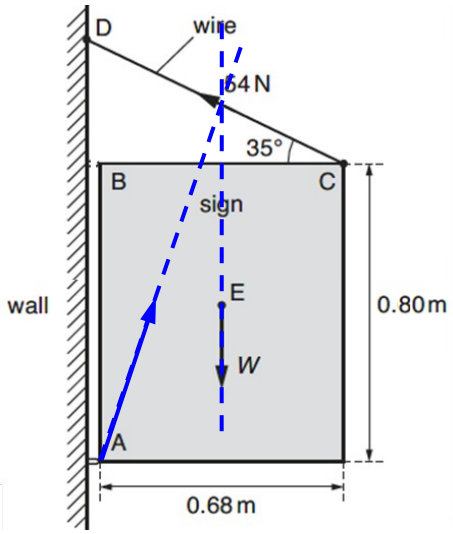
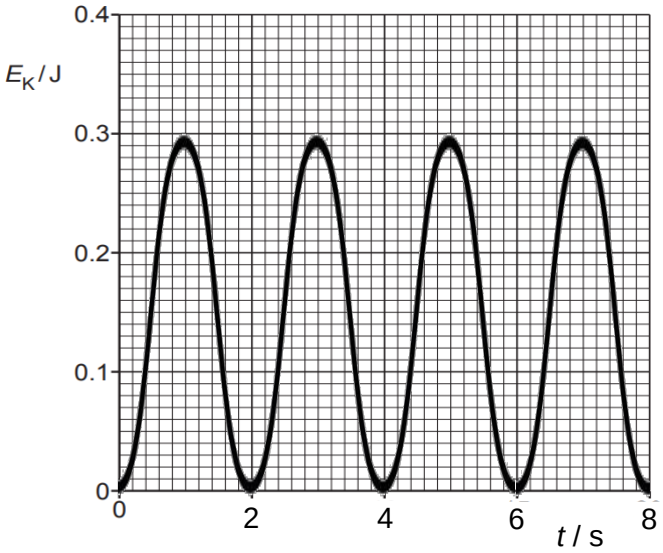
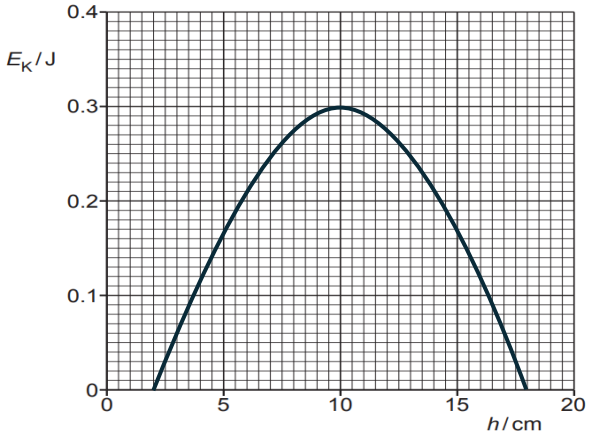
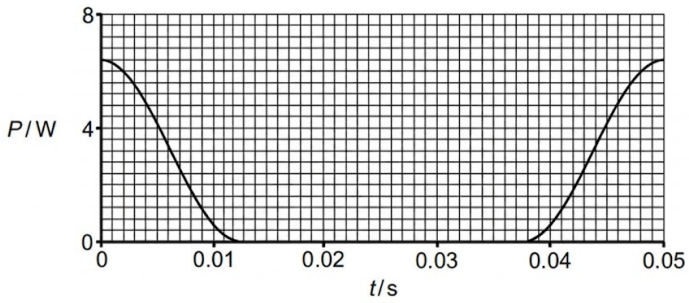
1	**(a)**	$E = \frac{4FL}{\pi d^2 x}$ $E = \frac{4 \times 25.0 \times 2.000}{\pi \times 0.00040^2 \times 1.20 \times 10^{-3}}$ $E = 3.3157 \times 10^{11}$	**B1**
		$E = \frac{4FL}{\pi d^2 x}$ $\frac{\Delta E}{E} = \frac{\Delta F}{F} + \frac{\Delta L}{L} + 2\frac{\Delta d}{d} + \frac{\Delta x}{x}$ $\frac{\Delta E}{3.3157 \times 10^{11}} = \frac{0.5}{25.0} + \frac{0.002}{2.000} + 2 \times \frac{0.02}{0.40} + \frac{0.06}{1.20}$	**M1**
		$\frac{\Delta E}{3.3157 \times 10^{11}} = 0.171$ $\Delta E = 5.6698 \times 10^{10}$ $\Delta E = 6 \times 10^{10}$ (1s.f.)	**M1** **A1**
		E = (3.3 ± 0.6) × 10^{11} N m^{-2} If E value is incorrect, the student can be awarded the two M1 marks for the method for calculating ΔE. No ECF for last A1 mark.	
	(b)(i)	The <u>resultant force</u> (acting on the sign) (in any direction) is <u>zero</u>. The <u>resultant torque or moments</u> <u>about any point</u> is <u>zero</u>.	**B1** **B1**
	(b)(ii)	(Since the force at point B is only in the vertical direction), The (line of action of the) <u>force at B</u> <u>passes through point A</u> or The force at B has <u>zero perpendicular distance</u> from point A.	 **B1**
	(b)(iii)	Using principle of moments, taking moments about A, Anti-clockwise moments about A = Clockwise moments about A $(54\sin 35^\circ \times 0.68) + (54\cos 35^\circ \times 0.80) = W \times 0.34$ W = 166 ≈ 170 N	 **C1** **A1**
	(b)(iv)	Considering vertical equilibrium $F_{Ay} + F_{By} + T_y = W$ $F_{Ay} + F_{By} + 54\sin 35^\circ = 166$ $F_{Ay} + F_{By} = 135 \approx 140\ N$	 **C1** **A1**

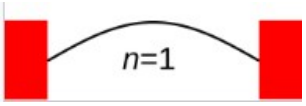
	(b)(v)	 For rotational equilibrium, the lines of action of all three non-parallel forces must intersect at a single common point. Thus the force at A should be acting towards the intersection of the lines of action of *W* and *T*.	**B1**
2	**(a)**	Energy is transferred from the kinetic store to electric potential store during approach. Energy is then transferred from electric potential store back kinetic store during separation. We can still accept: Kinetic energy (of the system) is converted to electric potential energy (of the system) during approach. Electric potential energy (of the system) is converted to kinetic energy (of the system) during separation.	**B1**
	(b)	Either The impulse/force they exert on each other has equal magnitude. The magnitude of the change in velocity of X is smaller than that for Y. $Ft = m \times \Delta v$, so X has larger mass. Or The impulse/force they exert on each other has equal magnitude. The gradient of X graph is smaller than the gradient of Y graph during the collision period. Or The impulse/force they exert on each other has equal magnitude.	**B1** **B1** **B1** **B1** **B1** **B1**

		The time for X to reach zero velocity is longer than the time for Y to reach zero velocity. The one with longer time will have a larger mass. Or The total momentum at the intersection point of the graph is positive. Hence the total momentum of the system must be positive. By principle of conservation of momentum, the total momentum at the start of collision must be positive. Since both particles have the same speed, X must have a larger mass.	**B1** **B1**
	(c)	From the graph, both particles reverse directions after collision. Take rightwards as positive, using conservation of momentum: $3mv + 2m(-v) = 3m(-v_x) + 2mv_y$ $v = 2v_y - 3v_x$ ..... (1) Relative speed of approach = relative speed of separation: $u_1 - u_2 = v_2 - v_1$ $v - (-v) = v_Y - (-v_X)$ $2v = v_Y + v_X$ ..... (2) (if student writes the conservation of kinetic energy equation, can award the second B1 mark) Sub v_y from (2) into equation (1): $v = 2(2v - v_x) - 3v_x$ $v_x = 0.6v$ Sub v_x into equation (2) $2v = v_Y + 0.6v$ $v_y = 1.4v$	**B1** **B1** **A1** **A1**
3	**(a)**	A periodic motion where the acceleration of the object is (directly) proportional to the displacement from the equilibrium position and always directed towards the equilibrium position. (allow: the direction of the acceleration is always opposite to the direction of the displacement.)	**B1** **B1**

	(b)(i)	$x_0 = \frac{1}{2}(18 - 2)$ $= 8.0$ cm	**A1**
	(b)(ii)	$T = 4.0$ s $\omega = \frac{2\pi}{4.0}$ $= 1.57 = 1.6$ rad s^{-1}	**M1** **A0**
	(b)(iii)	$E = E_{k\max}$ $E_{k\max} = \frac{1}{2}mv_o^2 = \frac{1}{2}m\omega^2 x_o^2$ $= \frac{1}{2}(37)(1.6)^2(8.0 \times 10^{-2})^2$ $= 0.303 = 0.30$ J Accept 0.29 J as student could have used ω = 1.57 rad s^{-1}	**C1** **A1**
	(c)	 1 mark for correct shape i.e. sine square shape 1 mark for correct timings for zero E_K i.e. E_K = 0 only at 0, 2 s, 4 s, 6 s, 8 s	**B1** **B1**
	(d)		

		1 mark for correct shape i.e. dome-shape curve with maximum E_K = 0.30 J and at position h = 10.0 cm 1 mark for correct intercepts i.e. E_K = 0 only at h = 2.0 cm and at h = 18 cm	**B1** **B1**
4	**(a)(i)**	(Since the power – time graph is symmetrical about the mean value,) Peak power dissipation P_0 = 6.4 W Mean power of ac source = ½ P_0 = 3.2 W	**A1**
	(a)(ii)	Mean power = i^2R $i = \sqrt{\frac{P}{R}} = \sqrt{\frac{3.2}{26}} = 0.35$ A	**A1**
	(b)(i)	 Correct readings - Peak power remains at 6.4W, no reading from 0.0125 to 0.0375 s Correct shape – cosine square function, straight zero line between 0.0125 to 0.0375 s	**B1** **B1**
	(b)(ii)	Since the diode allows only half of the AC supply to pass through, mean power = 3.2/2 = 1.6 W (Need to show the understanding that average power is halved the average power when not rectified) $i = \sqrt{\frac{P}{R}} = \sqrt{\frac{1.6}{26}} = 0.25$ A If student simply recall $I_{rms} = I_o / 2$ for half wave rectified, he will not get the first B1 mark.	**B1** **A1**

5	**(a)(i)**	$C = \frac{Q}{V}$ $= \frac{7.2 \times 10^{-3}}{12}$ $= 6.0 \times 10^{-4}$ F	**A1**
	(a)(ii)	$R = \frac{V}{I}$ $= \frac{12}{1.5 \times 10^{-3}}$ $= 8000\ \Omega$	**A1**
	(a)(iii)	$\tau = RC = 8000 \times 6.0 \times 10^{-4}$ $= 4.8$ s	**A1**
	(b)	V_C equals to V_R. V_C is (directly) proportional to charge Q on capacitor and V_R is (directly) proportional to current. Current is the rate of change of Q. [1] Therefore, Q is proportional to the rate of change of Q, hence Q has exponential relationship with t.	**B1** **B1**
	(c)(i)	$E = ½\ C\ V^2$ Therefore $E \propto V^2$ $\frac{E}{E_o} = (\frac{V}{V_o})^2$ $\frac{1}{2} = (\frac{V}{V_o})^2$ $V = \frac{1}{\sqrt{2}} V_o = 0.707 V_o$	**C1** **A1**

	(c)(ii)	For time T, i.e. when the voltage is halved, $\frac{1}{2}V_o = V_o e^{-\frac{t}{RC}}$ $\frac{T}{RC} = -\ln(\frac{1}{2})$ For time T', i.e. when energy is halved, the voltage is $\frac{1}{\sqrt{2}}V_o$, $\frac{1}{\sqrt{2}}V_o = V_o e^{-\frac{T'}{RC}}$ $\frac{T'}{RC} = -\ln(\frac{1}{\sqrt{2}})$ $T' = 0.50T$	**C1** **A1**
6	**(a)(i)**	The interference pattern shows that the electrons have wave-like nature since only waves can (diffract and) form interference pattern.	**B1**
	(a)(ii)	$qV = \frac{1}{2}mv^2$ $p = mv = \sqrt{2qVm}$	**M1** **A1**
	(a)(iii)	electrons have greater momentum so smaller (de Broglie) wavelength. Hence fringes become closer together.	**B1** **B1**
	(b)(i)	The <u>total probability</u> of finding the electron in the well is <u>one</u>.	**A1**
	(b)(ii)	Nodes at both walls, antinode at centre 	**A1**
	(b)(iii)	$E_1 = \frac{h^2}{8mL^2} = \frac{(6.63\times10^{-34})^2}{8\times(9.11\times10^{-31})(5.0\times10^{-9})^2}$ $E_1 = 2.4\times10^{-21}$ J	**A1**
	(b)(iv)	$E_2 = 4E_1 = 9.6\times10^{-21}$ J $\Delta E = E_2 - E_1 = 7.2\times10^{-21}$	**C1**

		$\lambda = \frac{hc}{\Delta E} = 2.8 \times 10^{-5}$ m	**A1**
7	**(a)**	Since equation is homogeneous $[W_0] = [c][v^2] = \text{kg m s}^{-2}$ $[c] = \frac{\text{kg m s}^{-2}}{(\text{m s}^{-1})^2}$ $= \text{kg m}^{-1}$	**A1** **A1**
	(b)(i)	$R = 2.3 + (0.6v^2 \times 10^{-3})$ $= 2.3 + (0.6)(80)^2 \times 10^{-3}$ $= 6.14$ Total resistive force = 6.14×300 = 1842 N = 1800 N	**C1** **A1**
	(b)(ii) 1.	80 km h^{-1} = 22.2 m s^{-1} Output power = $F \times v$ $= 42.5 \times 1000 \times 22.2$ $= 9.4 \times 10^5$ W	**C1** **A1**
	(b)(ii) 2.	Input power = $I \times V$ $= 2100 \times 750$ $= 1.6 \times 10^6$ W Efficiency $= \frac{9.4 \times 10^5}{1.6 \times 10^6} \times 100$ $= 59\%$	**C1** **A1**
	(b)(ii) 3.	The train draws power for the air-conditioning, communication system, lightings. If heat loss, student has to be specific on heat loss in the electrical motor instead of heat loss to surrounding. Award mark for any auxiliary systems related to the operation of the train other propulsion.	**A1**

	(c)(i)	$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{750}{22000}$ $= 0.034$	**C1** **A1**
	(c)(ii)	$\frac{N_s}{N_p} = \frac{I_p}{I_s}$ $\frac{750}{22000} = \frac{I_p}{7200}$ $I_p = 245$ A Peak current = $245 \times \sqrt{2}$ $= 347$ A	**C1** **A1**
	(d)(i)	KE of train = $\frac{1}{2}(300)(1000)(22.2)^2$ $= 7.4 \times 10^7$ J Amount of electrical energy produced = $0.30 \times 7.4 \times 10^7$ $= 2.2 \times 10^7$ J	**C1** **A1**
	(d)(ii)	Regenerative braking requires the power supply to be "receptive," meaning another train must be nearby to consume the energy. <u>During off-peak hours, there are fewer trains nearby.</u>	**A1**
	(e)(i)	$\beta = \frac{A_{tr}}{A_t}$ $0.65 = \frac{(3.2)(3.7)}{A_t}$ $A_t = 18$ m^2	**A1**
	(e)(ii)	Cause: As the train brakes, the air column trapped behind it continues to move forward due to its own momentum. Impact: The braking system must provide an additional force to overcome this forward air pressure to maintain the target deceleration.	**B1** **B1**
	(e)(iii)	Peak resistive drag during acceleration phase at t = 2.7 s When increasing the blockage ratio, Peak forward push during deceleration phase at t = 10.0 s	

When increasing the blockage ratio,

Min C_D (β = 0.65) = 3.0

Max C_D (β = 0.85) = 7.0

Increase in C_D = $\dfrac{(7.0 - 3.0)}{3.0} \times 100\% = 133\%$

	Max C_D value	Min C_D value	Percentage increase
Peak resistive drag during acceleration	11.8	7.2	64%
Peak forward push during deceleration	7.0 (6.5 to 7.5)	3.0 (2.5 to 3.5)	133%

A1

Allow tolerance in the reading.

Claim is supported because the percentage increase in C_D is significantly higher during the deceleration phase, i.e. impacting the braking system.

B1